

data stream that is also compatible with mobile sensors, even without extraneous interfaces. It provides a huge range of graphical interface tools, which enables the deployment of cloud based IoT applications, minimising the programming effort.

Open IoT is currently available as an open source project (<https://github.com/OpenIoTOrg/openiot/>). It also integrates libraries of the popular Global Sensor Networks (GSN) open source project. Recently, Open IoT was given the Black Duck award for being one of the top ten open source projects.

The OpenIoT platform

The Open IoT architecture comprises seven main elements.

1. The **sensor middleware (Extended Global Sensor Networks or X-GSN)** is firmware that provides us with the means to collect and analyse data streams. The sensor middleware is applied to the data to form the diverse administrating entities. The OpenIoT prototype implementation uses X-GSN (Extended GSN), an extended version of the GSN middleware. Besides, a mobile broker is used for integrating the mobile sensors with the middleware.
2. The **cloud data storage (Linked Stream Middleware Light, LSM-Light)** acts as a cloud database which ensures storage of large data streams coming up from the sensor middleware. The cloud infrastructure also supports working on metadata of all the data streams in OpenIoT. The OpenIoT prototype implementation uses the Linked Stream Middleware (LSM), which has been re-designed with push-pull data functionality and cloud interfaces.
3. The **scheduler process** requests for the on-demand provision of services, and ensures that it gets sustained resources for

OpenIoT capabilities

- Sensors and data streams registration, deployment and discovery
- Authenticated and authorised access to resources
- Zero-programming application development
- Handling of mobility with quality-driven sensor management

processing. It can use the services to find out resources that are compatible with the specific service.

4. The **service delivery and utility manager (SD&UM)** combines data streams as indicated by service workflows within the OpenIoT system in order to deliver the requested service (typically expressed as a SPARQL query). The SD&UM also acts as a service metering facility, which keeps track of utility metrics for each service.
5. The **request definition component** enables the on-the-fly specification of service requests to the OpenIoT platform. It comprises a set of services for specifying and formulating such requests, while also submitting them to the scheduler. This component is supported by a GUI (graphical user interface).
6. The **request presentation component** is in charge of imagining the rough idea of the outcome of all types of services. This component takes a few combinations of services
7. The **configuration and monitoring component** equips us with graphics management and the configuration of functionalities over services and sensors that are deployed within the OpenIoT platform.
OpenIoT has provided an innovative platform for IoT and cloud convergence, which enables:
 - Integration of IoT data and applications with cloud computing infrastructure
 - Deployment of and secure access to semantically interoperable applications
 - Handling of mobile sensors and associated QoS parameters
 The semantic interoperability functionalities of OpenIoT are a key differentiating factor of the project when compared to the wide range of other IoT and cloud platforms. These functionalities provide a basis for the development of novel applications in the areas of smart cities and mobile crowd sensing, while also enabling large scale IoT experimentation. 

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